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JEE MAIN PHYSICS DPP: UNITS & MEASUREMENTS

Topic: Systems of Units (CGS, FPS, MKS, SI), Fundamental & Derived Units | Time: 45 Min | Max Marks: 80

Marking Scheme: +4, -1 | Designed & Curated by: Syed Mazhar Ali

- Q1.** Which of the following sets represents quantities that are all fundamental base quantities in the International System of Units (SI)?
- Length, mass, velocity, luminous intensity
 - Mass, electric current, thermodynamic temperature, amount of substance
 - Electric charge, time, mass, luminous intensity
 - Length, force, time, temperature
- Q2.** In the CGS system, the unit of force is the dyne. The ratio of 1 N (SI unit of force) to 1 dyne is equal to:
- 10^3
 - 10^5
 - 10^7
 - 10^{-5}
- Q3.** The FPS system of units uses Foot, Pound, and Second. Given that 1 foot = 30.48 cm and 1 pound (mass) $\approx$ 453.6 g, the value of 1 slug (the standard FPS unit of mass defined as 1 lbf $\cdot$ s²/ft) in SI units (kg) is approximately:
- 0.4536 kg
 - 9.81 kg
 - 14.59 kg
 - 32.17 kg
- Q4.** In the SI system, plane angle and solid angle are:
- Fundamental quantities with units
 - Derived quantities without units
 - Supplementary/dimensionless derived quantities having units (radian and steradian)
 - Fundamental quantities without units
- Q5.** The SI unit of surface tension is N $\cdot$ m⁻¹. In the CGS system, its unit is dyne $\cdot$ cm⁻¹. A value of 0.072 N $\cdot$ m⁻¹ in the CGS system is:
- 7.2×10^{-4} dyne $\cdot$ cm⁻¹
 - 0.72 dyne $\cdot$ cm⁻¹
 - 72 dyne $\cdot$ cm⁻¹
 - 720 dyne $\cdot$ cm⁻¹
- Q6.** Which of the following is **not** a base unit in the SI system?
- Candela (cd)
 - Ampere (A)
 - Coulomb (C)
 - Mole (mol)
- Q7.** The universal gravitational constant G in SI units is 6.67×10^{-11} N $\cdot$ m² $\cdot$ kg⁻². Its numerical value in CGS units (dyne $\cdot$ cm² $\cdot$ g⁻²) is:
- 6.67×10^{-8}
 - 6.67×10^{-14}
 - 6.67×10^{-5}
 - 6.67×10^{-11}
- Q8.** The SI unit of dynamic viscosity η is Pa $\cdot$ s (kg $\cdot$ m⁻¹ $\cdot$ s⁻¹), whereas its CGS unit is poise (g $\cdot$ cm⁻¹ $\cdot$ s⁻¹). 1 Pa $\cdot$ s is equal to:
- 0.1 poise
 - 1 poise
 - 10 poise
 - 100 poise
- Q9.** If a new system of units is chosen where the unit of mass is 2 kg, the unit of length is 5 m, and the unit of time is 10 s, then 1 Joule of energy in this new system is equal to:
- 2 new units
 - 0.5 new units
 - 4 new units
 - 0.25 new units
- Q10.** Which of the following derived units is correctly expressed purely in terms of SI fundamental units?
- Pascal (Pressure) = kg $\cdot$ m $\cdot$ s⁻²
 - Watt (Power) = kg $\cdot$ m² $\cdot$ s⁻³
 - Volt (Potential) = kg $\cdot$ m² $\cdot$ s⁻² $\cdot$ A⁻¹
 - Tesla (Magnetic Field) = kg $\cdot$ m $\cdot$ s⁻² $\cdot$ A⁻¹
- Q11.** The density of a liquid is 13.6 g $\cdot$ cm⁻³ in CGS units. In a new system of units where the unit of mass is 100 g and the unit of length is 10 cm, the numerical value of the density is:
- 136
 - 13.6
 - 1360
 - 1.36
- Q12.** The heat capacity of a substance in SI units is expressed in J $\cdot$ K⁻¹. In terms of fundamental SI base units, it is written as:
- kg $\cdot$ m² $\cdot$ s⁻² $\cdot$ K⁻¹
 - kg $\cdot$ m $\cdot$ s⁻² $\cdot$ K⁻¹

- (c) $\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-3} \cdot \text{K}^{-1}$
 (d) $\text{kg} \cdot \text{m}^{-1} \cdot \text{s}^{-2} \cdot \text{K}^{-1}$

Q13. In a system where the unit of force is 100 N, the unit of length is 10 m, and the unit of time is 100 s, the unit of mass in this system corresponds to:

- (a) 10^3 kg
 (b) 10^4 kg
 (c) 10^5 kg
 (d) 10^2 kg

Q14. Which of the following physical quantities has a derived unit that is dimensionless in SI?

- (a) Permittivity of free space (ϵ_0)
 (b) Specific gravity (Relative density)
 (c) Universal gas constant (R)
 (d) Planck's constant (h)

Q15. In the CGS electrostatic unit (esu) system, the unit of electric charge is the statcoulomb. How many statcoulombs are there in 1 Coulomb of charge?

- (a) 3×10^8
 (b) 3×10^9
 (c) $1/(3 \times 10^9)$
 (d) 9×10^9

Q16. The SI unit of electrical resistance is the Ohm (Ω). Expressed entirely in base units (kg, m, s, A), 1Ω is equivalent to:

- (a) $\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-3} \cdot \text{A}^{-2}$
 (b) $\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-2} \cdot \text{A}^{-2}$
 (c) $\text{kg} \cdot \text{m} \cdot \text{s}^{-3} \cdot \text{A}^{-2}$
 (d) $\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-3} \cdot \text{A}^{-1}$

Q17. The value of acceleration due to gravity is $g = 9.8 \text{ m} \cdot \text{s}^{-2}$. If the unit of length is chosen to be 1 km and the unit of time is 1 minute, the numerical value of g in this new system will be:

- (a) 35.28
 (b) 3.528
 (c) 352.8
 (d) 0.3528

Q18. The SI unit of inductance is Henry (H). Which of the following combinations of base units correctly represents 1 H?

- (a) $\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-2} \cdot \text{A}^{-2}$
 (b) $\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-1} \cdot \text{A}^{-2}$
 (c) $\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-2} \cdot \text{A}^{-1}$
 (d) $\text{kg} \cdot \text{m} \cdot \text{s}^{-2} \cdot \text{A}^{-2}$

Q19. The work done in moving a body is 10 J in MKS/SI. If work is to be converted into CGS (erg), 10 J corresponds to:

- (a) 10^6 erg
 (b) 10^7 erg
 (c) 10^8 erg
 (d) 10^5 erg

Q20. The Stefan-Boltzmann constant σ has the SI unit $\text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}$. In terms of SI base fundamental units, it is:

- (a) $\text{kg} \cdot \text{s}^{-3} \cdot \text{K}^{-4}$
 (b) $\text{kg} \cdot \text{m} \cdot \text{s}^{-3} \cdot \text{K}^{-4}$
 (c) $\text{kg} \cdot \text{s}^{-2} \cdot \text{K}^{-4}$
 (d) $\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-3} \cdot \text{K}^{-4}$

ANSWER KEY & STEP-BY-STEP SOLUTIONS

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Quick Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(b)	6	(c)	11	(a)	16	(a)
2	(b)	7	(a)	12	(a)	17	(a)
3	(c)	8	(c)	13	(c)	18	(a)
4	(c)	9	(a)	14	(b)	19	(c)
5	(c)	10	(b)	15	(b)	20	(a)

Detailed Solutions

Sol 1. Correct Option: (b)

The 7 SI fundamental base quantities are Length (m), Mass (kg), Time (s), Electric Current (A), Thermodynamic Temperature (K), Amount of Substance (mol), and Luminous Intensity (cd). Velocity, force, and electric charge ($Q = It$) are derived quantities.

Sol 2. Correct Option: (b)

Force = Mass $\times$ Acceleration.

$$1 \text{ N} = 1 \text{ kg} \cdot \text{m} \cdot \text{s}^{-2}.$$

$$1 \text{ dyne} = 1 \text{ g} \cdot \text{cm} \cdot \text{s}^{-2}.$$

Conversion:

$$1 \text{ N} = (10^3 \text{ g}) \times (10^2 \text{ cm}) \times \text{s}^{-2} = 10^5 \text{ dyne}$$

$$\text{Thus, } \frac{1 \text{ N}}{1 \text{ dyne}} = 10^5.$$

Sol 3. Correct Option: (c)

By definition in the FPS engineering system, a slug is the mass that accelerates at $1 \text{ ft} \cdot \text{s}^{-2}$ under a force of 1 lbf (32.174 lb mass):

$$1 \text{ slug} = 32.174 \text{ lb} \approx 32.174 \times 0.45359 \text{ kg} \approx 14.59 \text{ kg}$$

Sol 4. Correct Option: (c)

Plane angle ($\theta = l/r$, unit: radian) and solid angle ($\Omega = A/r^2$, unit: steradian) are dimensionless quantities (ratio of two lengths or two areas), but they possess distinct supplementary units in SI.

Sol 5. Correct Option: (c)

$$\text{Surface tension } T = \frac{F}{L}.$$

$$1 \text{ N} \cdot \text{m}^{-1} = \frac{10^5 \text{ dyne}}{10^2 \text{ cm}} = 10^3 \text{ dyne} \cdot \text{cm}^{-1}.$$

Therefore:

$$0.072 \text{ N} \cdot \text{m}^{-1} = 0.072 \times 10^3 = 72 \text{ dyne} \cdot \text{cm}^{-1}$$

Sol 6. Correct Option: (c)

Coulomb (C) is a derived unit of electric charge, defined as $\text{A} \cdot \text{s}$. The base unit in SI for electricity is the Ampere (A).

Sol 7. Correct Option: (a)

$$G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2 \cdot \text{kg}^{-2}.$$

$$\text{Using } n_2 = n_1 \left[\frac{M_1}{M_2} \right]^{-1} \left[\frac{L_1}{L_2} \right]^3 \left[\frac{T_1}{T_2} \right]^{-2}:$$

$$n_2 = 6.67 \times 10^{-11} \times \left(\frac{1000 \text{ g}}{1 \text{ g}} \right)^{-1} \times \left(\frac{100 \text{ cm}}{1 \text{ cm}} \right)^3$$

$$n_2 = 6.67 \times 10^{-11} \times 10^{-3} \times 10^6 = 6.67 \times 10^{-8}$$

Sol 8. Correct Option: (c)

$$1 \text{ Pa} \cdot \text{s} = 1 \text{ kg} \cdot \text{m}^{-1} \cdot \text{s}^{-1}.$$

Conversion to CGS:

$$1 \text{ Pa} \cdot \text{s} = \frac{1000 \text{ g}}{(100 \text{ cm}) \cdot \text{s}} = 10 \text{ g} \cdot \text{cm}^{-1} \cdot \text{s}^{-1} = 10 \text{ poise}$$

Sol 9. Correct Option: (a)

Energy has dimensions $[M^1 L^2 T^{-2}]$.

$$\text{Using conversion formula } n_2 = n_1 \left(\frac{M_1}{M_2} \right)^1 \left(\frac{L_1}{L_2} \right)^2 \left(\frac{T_1}{T_2} \right)^{-2}$$

$$n_2 = 1 \times \left(\frac{1 \text{ kg}}{2 \text{ kg}} \right)^1 \times \left(\frac{1 \text{ m}}{5 \text{ m}} \right)^2 \times \left(\frac{1 \text{ s}}{10 \text{ s}} \right)^{-2}$$

$$n_2 = 1 \times \frac{1}{2} \times \frac{1}{25} \times 100 = \frac{100}{50} = 2 \text{ new units}$$

Sol 10. Correct Option: (b)

$$\text{Power} = \frac{\text{Work}}{\text{Time}} = \frac{\text{Force} \times \text{Distance}}{\text{Time}}.$$

Base units:

$$\text{Watt} = \frac{(\text{kg} \cdot \text{m} \cdot \text{s}^{-2}) \cdot \text{m}}{\text{s}} = \text{kg} \cdot \text{m}^2 \cdot \text{s}^{-3}$$

$$\text{Pascal} = \text{kg} \cdot \text{m}^{-1} \cdot \text{s}^{-2}, \text{ Volt} = \text{kg} \cdot \text{m}^2 \cdot \text{s}^{-3} \cdot \text{A}^{-1},$$

$$\text{Tesla} = \text{kg} \cdot \text{s}^{-2} \cdot \text{A}^{-1}.$$

Sol 11. Correct Option: (a)

Density has dimensions $[M^1 L^{-3}]$.

$$n_1 = 13.6, M_1 = 1 \text{ g}, L_1 = 1 \text{ cm}.$$

$$M_2 = 100 \text{ g}, L_2 = 10 \text{ cm}.$$

$$n_2 = 13.6 \times \left(\frac{1}{100} \right) \times \left(\frac{1}{10} \right)^{-3} = 13.6 \times \frac{1000}{100} = 136$$

Sol 12. Correct Option: (a)

$$\text{Heat Capacity } C = \frac{Q}{\Delta T}.$$

Since Q is energy ($\text{J} = \text{kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$) and ΔT is in Kelvin (K):

$$[C] = \text{kg} \cdot \text{m}^2 \cdot \text{s}^{-2} \cdot \text{K}^{-1}$$

Sol 13. Correct Option: (c)

Given: Force $F = 100 \text{ N}$, Length $L = 10 \text{ m}$, Time $T = 100 \text{ s}$.

Since $F = MLT^{-2}$, we solve for unit mass M :

$$M = \frac{FT^2}{L} = \frac{100 \times (100)^2}{10} = 10^5 \text{ kg}$$

Sol 14. Correct Option: (b)

Specific gravity is the ratio of the density of a substance to the density of water at 4°C . Being a ratio of two identical quantities, it is dimensionless and unitless.

Sol 15. Correct Option: (b)

From electrostatics in CGS-esu vs SI:

1 Coulomb = 3×10^9 statcoulomb (or esu of charge).

Sol 16. Correct Option: (a)

Resistance $R = \frac{V}{I} = \frac{W}{Q \cdot I} = \frac{\text{Energy}}{I^2 \cdot t}$.

In SI base units:

$$1 \Omega = \frac{\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-2}}{\text{A}^2 \cdot \text{s}} = \text{kg} \cdot \text{m}^2 \cdot \text{s}^{-3} \cdot \text{A}^{-2}$$

Sol 17. Correct Option: (a)

Acceleration has dimensions $[L^1 T^{-2}]$.

$n_1 = 9.8$, $L_1 = 1 \text{ m}$, $T_1 = 1 \text{ s}$.

$L_2 = 1 \text{ km} = 1000 \text{ m}$, $T_2 = 1 \text{ min} = 60 \text{ s}$.

$$n_2 = 9.8 \times \left(\frac{1}{1000}\right) \times \left(\frac{1}{60}\right)^{-2} = 9.8 \times 3.6 = 35.28$$

Sol 18. Correct Option: (a)

Energy stored in an inductor $U = \frac{1}{2}LI^2 \implies$

$$L = \frac{2U}{I^2}.$$

Units of energy = $\text{J} = \text{kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$.

Units of $I^2 = \text{A}^2$.

Therefore:

$$1 \text{ H} = \text{kg} \cdot \text{m}^2 \cdot \text{s}^{-2} \cdot \text{A}^{-2}$$

Sol 19. Correct Option: (c)

Work in SI: $1 \text{ J} = 1 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-2} = 10^3 \text{ g} \times 10^4 \text{ cm}^2 \times \text{s}^{-2} = 10^7 \text{ erg}$.

For 10 J:

$$10 \text{ J} = 10 \times 10^7 \text{ erg} = 10^8 \text{ erg}$$

Sol 20. Correct Option: (a)

Stefan-Boltzmann law: $E = \sigma T^4 \implies \sigma = \frac{E}{T^4}$.

Power per unit area: $\frac{\text{W}}{\text{m}^2} = \text{kg} \cdot \text{s}^{-3}$.

Dividing by K^4 :

$$[\sigma] = \text{kg} \cdot \text{s}^{-3} \cdot \text{K}^{-4}$$

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